Akbar Abilov

F-1 OPT, STEM OPT eligible | aa13471@nyu.edu | github.com/ekberby

Education

New York University – Master of Science in Urban Data Science — Expected May 2027
University of Essex – Bachelor of Science in Computer Science — July 2025

Experience

Data Platform Developer, NYU Marron Institute of Urban Management – New York, NY — Oct 2026 – Present

- Architected a 10-module system and relational data model that unifies case management, project management, and monitoring and evaluation for field programs handling sensitive personal data, replacing Excel-based tracking
- Translated 47 stakeholder requirements and 6 use cases into a design specification, tracing 90 design changes to their source and closing 3 requirement gaps found in a completeness check
- Designed offline data sync and role-based access control with consent checks, so field staff can record data without connectivity while personal records stay protected

BIM Automation Intern, PortBIM — May 2025 – Aug 2025

- Reduced manual engineering workflow time by 60–70% by building 5+ automation tools for Autodesk Revit and AutoCAD in C# and Python
- Improved automation speed 2–3x by refactoring Autodesk Dynamo visual scripts into compiled C# code
- Saved engineers 10+ hours per week by building a centralized update and configuration system that deployed new tool versions to every workstation at once

Projects

Pull, Congressional Bill Tracker — github.com/NYU-Congressmen/congress-tracker

- Recovered 90 of 19,832 bills silently dropped on every sync by tracing unstable offset paging over 2,023 tied sort dates and adding a count-checked re-read with deduplication in Python
- Reduced bill summary sync from about 19,700 API requests to 26 by switching to a date-ranged bulk endpoint, loading all 6,446 summaries within a 5,000-request hourly limit
- Designed 14 PostgreSQL tables across 3 Alembic migrations with CHECK, UNIQUE, and foreign key constraints, and built FastAPI search endpoints tested with pytest in GitHub Actions, in a 4-person team

AskNYC Zoning Assistant — github.com/ekberby/AskNYC

- Built a retrieval-augmented generation (RAG) assistant in Python that cites a zoning section for every claim, retrieving the top 3 section-tagged chunks through FAISS and Sentence-Transformers semantic search
- Packaged the app in Docker with a locally hosted Llama 3.1 model (Ollama) and a Streamlit interface, so it runs with 1 command and returns answers with sources as structured JSON

Trip Management Web App — github.com/ekberby/Tourism_project_web_demo

- Built a full-stack Java Spring Boot web app with registration, login, and trip management on a layered controller, service, and repository architecture over PostgreSQL, securing every page with Spring Security

NYC Bus Speed Prediction

- Predicted bus speed at R^2 0.704 (vs 0.435 baseline) and severe delays at ROC-AUC 0.900 by merging 5 transit datasets into a 6.14M-row pandas table and training scikit-learn random forests in a 4-person team

Satellite Land-Use Classification — github.com/ekberby/Deep-Learning-Land-Use

- Reached 92.6% test accuracy on 27,000 EuroSAT satellite images across 10 land-use classes, with 97–99% precision on forest and water, by fine-tuning a pretrained ResNet18 in PyTorch

Skills

Languages: Python, Java, SQL, C#
Frameworks and Tools: FastAPI, Spring Boot, PostgreSQL, Alembic, pytest, Docker, GitHub Actions, Streamlit
Libraries and Software: pandas, scikit-learn, PyTorch, FAISS, Sentence-Transformers, Autodesk Revit, AutoCAD